

Nonnegative and positive factorizations

Full nonnegative rank of the quadratic correlation matrix

Difficulty: extreme **Importance:** broadly interesting

Status: [Lean verified](#)

Rating rationale: Historical ratings for the original conjecture. Extreme because full nonnegative rank is much sharper than available exponential lower bounds for this family; broad importance concerns communication complexity and limitations of linear programming formulations.

Area: exact NMF and lower bounds for optimization formulations

Negative resolution — 2026-09-13

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. **Independent Codex AI-agent informal audit:** PASS for the complete original target.

[Theorem 1](#), proved in [Sections 2–4](#) constructs nonnegative rational factors for the fully prescribed matrix and proves

$$\text{rank}_+(C_n) \leq \min \left\{ 2^n, 2^{n-1} + n + \binom{n}{2} + \binom{n}{4} \right\}, \quad n \geq 1.$$

In particular, $\text{rank}_+(C_7) \leq 127 < 128$, so the original equality for every $n \geq 3$ is false. The bound is strictly below 2^n for every $n \geq 7$. All entries remain fixed by the original formula, over the original real field; positive column denominators convert the exact integer certificate into genuine rational factors. This is a complete negative resolution of the universal question. Exact ranks at $n = 5, 6, 7$ and the smallest counterexample dimension are not determined or required, and no extension-complexity claim for the whole correlation polytope is made.

[Standalone proof source](#) · [Independent review](#) · [Exact certificate](#) · [Submission, verified affiliation and reproduction](#).

The complete argument passed independent informal AI-agent review, with fresh exact checks. ChatGPT assistance is disclosed; neither external human peer review nor formal verification is claimed. That informal audit did not assert formal verification; the separate Lean verification below covers the $n=7$ counterexample contracts. The original statement and prior partial-result credit are retained below.

Historical partial result — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the stated partial scope.

The fixed three-bit quadratic correlation matrix has nonnegative rank exactly eight. Its ordinary-rank-seven parity null vector constrains both factors in a hypothetical seven-term factorization; nine distinguished entries then exclude such a factorization.

Remaining question at that date: The full prescribed-completion conjecture for every $n \geq 4$ remained unresolved. The parity restriction is proved only for a hypothetical factorization whose inner dimension equals ordinary rank; it is not imposed on arbitrary wider factorizations. The new counterexample above now settles the universal conjecture negatively; these earlier findings retain their original credit.

Primary reference: [complete authored PDF](#), [standalone TeX](#), [Theorem 1 and Lemma 2](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Lean proof and verification evidence — 2026-09-13

Formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA. The complete mathematical counterexample and exact factorization are by Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. Matthew J. Colbrook’s earlier partial result remains separately credited.

The [immutable complete Lean proof](#) uses Lean 4.33.1, [LeanCert 621a43d7cf21](#) and [Mathlib 0df444a360ea](#), with all dependency pins in the [manifest](#). The [project guide](#), [numerical targets](#), [definitions](#) and [ten-contract boundary](#) record the exact statement correspondence.

The export `NLA.NR03.not_targetStatement` negates the complete displayed universal equality at the prescribed $n = 7$ matrix. The public definition continues to quantify all $n \geq 3$, every Boolean row and column, and the real square $(1 - \text{boolDot } a \ b)^2$; the formal proof supplies the complete width-127 counterexample and the bridges from its integer certificate to genuine real nonnegative factors. It does not claim exact ranks at $n = 5, 6, 7$, that 7 is the smallest counterexample, or a statement about the full correlation-polytope slack matrix.

The [complete canonical Linux run](#) verified all ten exported statements, LeanCert kernel trust, Comparator statement correspondence and Lean’s default-kernel replay. The same job passed sandbox and rejection controls. Every public theorem reports exactly the standard axioms *propext*, *Classical.choice* and *Quot.sound*; the retained raw evidence binds all 113 input files to proof revision `f664d07e82aaa60bc9c78dd1946e763168c5c530`.

Two independent final AI-agent reviews are retained at [final referee 1](#) and [final referee 2](#). They inspect the final proof source and exact target; they are not external human peer review and do not claim official Tau Ceti endorsement. The formalization author, source authors and historical partial credit remain exactly as identified above.

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

For each integer $n \geq 3$, let C_n be the $2^n \times 2^n$ matrix indexed by $a, b \in \{0, 1\}^n$ and defined by

$$C_n(a, b) = (1 - a^\top b)^2.$$

Question

Is $\text{rank}_+(C_n) = 2^n$ for every $n \geq 3$? All entries, including those for $a^\top b > 1$, are fixed by this formula.

The matrix has ordinary rank $1 + n(n + 1)/2$ and is a submatrix of a slack matrix of the correlation polytope formed from valid, possibly redundant inequalities. The conjecture therefore asks for a large separation between ordinary and nonnegative matrix rank with consequences for linear programming representations.

References

Vandaele, Gillis, Glineur, and Tuytens, [Heuristics for Exact Nonnegative Matrix Factorization](#), §6.4, Conjecture 4. Gillis, [Nonnegative Matrix Factorization](#), §3.7, p. 96, using the name U_n for this fixed matrix.

Status check — 2026-09-10

Rechecked [Baeckelant et al. v2, §6.7 and Appendix A.4](#) and [Sergeev's July 2026 paper](#), and searched for a full-rank resolution. The former retains the prescribed-completion conjecture, including the unresolved $n=3$ case. Sergeev treats a partial unique-disjointness matrix whose remaining entries may vary; this does not determine the rank of the fixed matrix here. No full resolution was located.